

Guardian: Real-Time AI Scam Protection for India's Next Billion Users

Bhashini API Key Request

Hackindore Submission — Bhashini Domain Innovation Challenge 1.0

EXECUTIVE SUMMARY

Guardian is a multilingual, AI-powered scam protection app for Indian mobile users. It protects users across the entire scam kill chain — SMS, WhatsApp notifications, live phone calls, and UPI payment moments — and warns them in their own language, in real time, before money leaves their account.

Built on Bhashini's language AI stack (Streaming ASR, Translation, TTS) and Google's Gemini 1.5 Flash for zero-shot scam reasoning, Guardian is designed for India's linguistic reality: Hindi, English, Hinglish, and 22 scheduled languages.

We are requesting a Bhashini API key to power real-time Indian-language transcription, bidirectional translation, and voice warnings for low-literacy users.

Core Project Mission & Architectural Overview

Digital financial fraud in India increasingly exploits linguistic diversity and psychological coercion. Guardian bridges the gap between state-of-the-art foundation AI models and vernacular speakers across rural and urban India.

Pillar	Specification & Technical Implementation
Target Audience	Elderly, rural, first-time smartphone users, and regional language speakers.
Attack Surfaces	Inbound phone calls (digital arrest, KYC fraud), SMS/WhatsApp phishing, QR codes, & UPI triggers.
Language Stack	Bhashini IndicTrans2, Streaming ASR (VAD WebSocket), and Indic TTS.
AI Reasoning	Gemini 1.5 Flash (five-engine cognitive model) with pure Kotlin offline keyword fallback.
Interception Method	Default Browser role, NotificationListenerService, Speakerphone ASR listener, & Floating HUD.

The Problem

India loses over ₹1,750 crore to cyber fraud every year. The majority of victims are elderly, rural, and non-English speaking. Scams arrive through SMS, WhatsApp, phone calls, and UPI — and AI-generated voices make them harder to detect.

Existing solutions fail:

- Truecaller: only flags known numbers; misses new scam scripts
- Carrier filters: only detect bulk SMS; miss targeted attacks
- Bank warnings: arrive after money is gone
- Google Safe Browsing: only covers links, not calls or SMS

The gap: No existing tool protects Indian users in their own language across all scam channels in real time.

The Solution

Guardian runs silently on the user's phone and warns them before they act.

Multi-Channel Protection:

- SMS / WhatsApp — NotificationListenerService reads messages before user taps
- Phone Calls — Live call audio transcribed via Bhashini ASR
- QR Codes — CameraX + ML Kit scans before payment
- Links — Intercepts via default browser role
- UPI Moment — Warns if user opens a payment app during a flagged call

The Bhashini Pipeline:

```
User speaks (Hindi / Hinglish / English)
↓
Bhashini Streaming ASR (WebSocket, VAD-enabled)
↓
Live transcript in user's language
↓
Bhashini Translation → English
↓
Gemini 1.5 Flash — five-engine semantic analysis
↓
RiskReport with explanation_en + explanation_hi
↓
Bhashini Translation → back to user's language
↓
Floating AI Reasoning Card (in user's language)
```

This pipeline lets an English-trained LLM reason about Hindi speech and delivers the explanation back in the user's mother tongue.

Five-Engine Semantic AI:

1. Pretext Legitimacy — Is the caller faking a bank / police / courier identity?
2. Intent & Action Risk — Are they trying to extract OTP, PIN, or funds?
3. Psychological Pressure — Are they manufacturing fear or urgency?
4. Information Asymmetry — Are they demanding secrets while giving vague claims?
5. Zero-Shot Variant Detection — Catches new scam scripts never seen before

The AI Reasoning Card: The user sees a live floating overlay during the call: Risk score (0–100%) with color gradient, signal breakdown, tactical explanation in Hindi/English, and highlighted risky phrases.

Family Protection Layer: When a scam is detected, family members receive a push notification with caller details and a 'Call them now' button. One-tap reporting to cybercrime.gov.in is included.

Bhashini APIs We Will Use

- Streaming ASR (WebSocket) — Real-time transcription in Hindi, English, Tamil, Telugu, Bengali, and 19 other languages with built-in VAD and code-mixed speech support
- Translation API (IndicTrans2) — Bidirectional translation: user language → English for LLM, English → user language for display
- Text-to-Speech (TTS) — Voice warnings for visually impaired or low-literacy users

Technical Stack

- Client (Android): Kotlin, Jetpack Compose, CameraX, ML Kit, NotificationListenerService, Foreground Service, EncryptedSharedPreferences
- Bhashini Integration: Streaming ASR via WebSocket (OkHttp), Translation via REST (Retrofit), TTS via REST
- AI Analysis: Google Gemini 1.5 Flash (five-engine analysis), local keyword fallback for offline mode
- Backend: Node.js + Express, Firebase Cloud Messaging for family alerts

Why Bhashini Is Essential

Without Bhashini, Guardian would work only in English — and English-only scam protection is useless for the majority of Indians who are targeted. Bhashini's ASR, Translation, and TTS infrastructure is what turns Guardian from an English app into a genuinely national product.

Running on India's public digital infrastructure means:

- Data stays in India
- Free developer tier makes it accessible to small teams
- Directly advances the National Language Translation Mission
- Future-proofs against commercial vendor lock-in

Expected Impact

Metric	Target Projection
Scam events flagged per user per week	3 – 5 threats blocked
Time from call start to first alert	< 3 seconds (low latency)
Flagged items confirmed as scams	> 70% precision rate
Family alerts per active circle	1 – 2 per month
Cybercrime reports filed via Guardian	500+ in first 6 months

Honest Limitations

- Android blocks third-party apps from directly accessing in-call audio. Guardian uses the speakerphone + microphone workaround — the same approach used by accessibility apps like Rogervoice.
- AI-voice detection on compressed phone audio is an unsolved problem. Guardian provides a confidence score, not a guarantee.
- Link interception requires setting Guardian as the default browser — an Android platform constraint.

About the Team & Contact

Guardian is being built for Hackindore under the Bhashini Domain Innovation Challenge 1.0 track. The team combines Android development, AI integration, and product design experience.

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